Midterm Review

Modules 1–4: optimization, sparsity, kernels and SVMs, then PCA through tensors.

Week 9 has no lecture slides; this page is a review checklist. The midterm is **Modules 1–4** (weeks 1–8). Canvas has the official time and room.

What should be in your hands, not in a formula sheet you have never used:

- Gradients and Hessians; first- and second-order tests; convexity.
- Line search, GD, momentum, SGD, Newton / BFGS. Condition numbers.
- Lagrangians, a small KKT system, LP and QP duals.
- Ridge versus LASSO (Module 2). Soft thresholding in one sentence.
- Kernel trick, kernel ridge, primal and dual SVM (Module 3).
- PCA / SVD, kernel PCA, CCA or IVA at the level of "what is being correlated," tensors as multi-way arrays (Module 4).

Quizzes and exams stay on Canvas, not on this page.

Bring questions to office hours. Write one derivation you still cannot do with the notes closed, and do it.